MASTER OF SCIENCE
IN ENGINEERING

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Advanced Statistical Data Analysis
Worksheet Week 2

Exercise 1: Sniffer Data When gasoline is pumped into a tank, hydrocarbon vapours are forced out of the tank and into the atmosphere. To reduce this significant source of air pollution, devices are installed to capture the vapour. In testing these vapour recovery systems, a "sniffer" measures the amount recovered. To estimate the efficiency of the system, some method of estimating the total amount given off must be used. To this end, a laboratory experiment was conducted in which the amount of vapour given off was measured under controlled conditions. Four predictors are relevant for modelling:

`Temp.Tank`	=	initial tank temperature (degrees F)
`Temp.Gas`	=	temperature of the dispensed gasoline (degrees F)
`Pres.Tank`	=	initial vapour pressure in the tank (psi)
`Pres.Gas`	=	vapour pressure of the dispensed gasoline (psi)

The response is the hydrocarbons `Y` emitted in grams. The data, are given in the data file `sniffer.dat`

a) Fit the regression model (in `R` notation)

`Y ~ Temp.Tank + Temp.Gas + Vapor.Tank + Vapor.Dispensed`

by the least squares method.

- Report the estimated coefficients.
- What do you conclude when considering the marginal P-values and the p-value of the "F-statistic" in the summary output?
- Determine the variance inflation factors for all of the coefficients. What do you conclude from these results?
- Does the model adequately describe the data? Conduct a residual analysis.

b) Start from the model fitted in part (a) and perform a variable selection using the AIC stepwise. Report and discuss the result.

c) We detected strong multicollinearity in the initial regression model (cf. part (a)). Have we remedied these deficiencies by the variable selection?
If not, modify the predictors / regression model to fix it. Are you able to interpret the estimated parameters appropriately in the fixed regression model? Does this model still adequately describe the data?

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Exercise 2: The data set `Jet.dat` contains data on the trust, `Y`, of a jet turbine engine. Fit a regression model to these data using all the predictors

`x1`: Primary speed of rotation	`x4`: Pressure
`x2`: Secondary speed of rotation	`x5`: Exhaust temperature
`x3`: Fuel flow rate	`x6`: Ambient temperature at time of test

Use variable transformation, variable selection and multicollinearity analysis to find an appropriate regression model for these data. Investigate model adequacy using residual and sensitivity plots.

Exercise 3: Continue Exercise 2 from Worksheet Week 1 (power generating windmills).

a) Predict the `DC.output` for wind velocities of one and ten meter per second using the fitted regression model

$$\text{DC.output}_i = \beta_0 + \beta_1 \frac{1}{\text{velocity}_i} + E_i \quad \text{with } E_i \text{ indep.} \sim \mathcal{N}\left\langle 0, \sigma^2 \right\rangle .$$

and find a 95% prediction interval at both velocities. Please comment your results.

Exercise 4: Continue Exercise 3 from Worksheet Week 1 (Construction of LWR plants).

Remember: The data was collected with the aim of predicting the cost of construction of further LWR plants. Six of the power plants had partial turnkey guarantees and it is possible that, for these plants, some manufacturers' subsidies may be hidden in the quoted capital costs. Thus, we are interested in how this effect affects the costs of construction adjusted by the other influencing effects.

a) Find a most parsimonious regression model using the R function `step()` starting from the model fitted in Exercise 3 of Worksheet Week 1.

b) Assess the quality of the selected regression model by a residual and sensitivity analysis.

c) Answer the question in the introductory text.

d) Is there a multicollinearity issues in the resulting model of (a)?

Exercise 5: Reproduce the analysis of Henderson and Velleman as given in the Lecture Note 2.3.c. Assess the fit by a residual and sensitivity analysis. Compare the final model with "our" model of Section 2.2 (cf. Lecture Note 2.3.d).

You find the corresponding R code in the file `AdvStDaAn_Chap2withR.R`.